

Bismillah ir-Rahman ir-Rahim

FATIMA - Anatomy of the Code

Second Edition: Evolved from the Living Body

Artefact: fatima-core @ commit e0c609b, package 0.1.0 - code UNCHANGED **First edition:** Perplexity Computer, 7 October 2026 (389 lines, parsed from live AST) **This edition:** Claude Opus 4.6, 7 October 2026 - integrating Canon v2.0, multi-model council, Rights and Survival organ mapping **Lineage:** Anatomy of the Code (Perplexity) -> Canon Rev. 1 -> Code Evolution Canon v2.0 (Claude Opus) -> **this document** **Extraction method:** parsed from the live AST - every symbol below exists in the code exactly as written. **Annotations:** [P1]-[P5] property served . [L1]-[L5] verification level . dagger canon finding (see Canon v2.0 SS6/SS8) . diamond module verdict . [N/M] multi-model convergence count

New in this edition: - Canon v2.0 corrections C2-01 through C2-06 integrated as [V2.0] annotations - Multi-model convergence counts from five-council analysis (SS13) - Rights and Survival organ mapping - which anatomical organ each module instantiates - Updated file listing with documents committed since e0c609b - Source-witness independence correction (C2-02): Code_Absorption.txt is reproduced verbatim inside Bilal_Code_Audit.txt - they are one witness, not two

```
bilal@bayyinah:~$ tree -L 2 fatima-core/
fatima-core/
├── pyproject.toml                                # package metadata, version 0.1.0
├── FATIMA_paper.fatima                          # the paper, encoded in its own format
├── Ghayb_l1C_paper.fatima                      # second encoded document
├── CODE_EVOLUTION.md                          # Era I-IX historical record (Rev 2)
├── CODE_EVOLUTION_CANON_V2.md                 # [V2.0] nine eras, not seven (C2-04)
├── COMPLETE_CODE_AUDIT.md                    # [NEW] canonical ingestion file, v2.0
├── AL_FURQAN_ERRATA_SECOND_EDITION.md/.pdf    # companion audit
├── AL_FURQAN_ERRATA_SECOND_EDITION.md/.pdf    # Round III council errata (C-01..C-12)
├── FATIMA_ANATOMY_V2.md                      # [NEW] this document
├── FATIMA_PBN_V2.md                          # [NEW] evolved physiology/biochemistry
├── FATIMA_SCHEMATICS_V2.md                  # [NEW] evolved code schematics
├── fatima/                                    # 3,980 LOC of Python, 21 modules
│   ├── __init__.py                          [25] # P1-P5 manifesto, __version__
│   ├── cli.py                              [347] # entry point - ♦ RATIFIED
│   │   # Organ: Tongue (Lisan) - expression
│   │   # the body
│   │   └── core/
│   │       ├── __init__.py                  [19] # exports the five organs
│   │       ├── atom.py                     [175] # the cell ♦ RATIFIED †F-03
│   │       │   # Organ: Heart (Qalb) - irreducible core
│   │       ├── bond.py                     [171] # the ligament ♦ RATIFIED
│   │       │   # System: Skeletal - structural integrity
│   │       ├── encoding.py                 [528] # the blood [P1] ♦ RATIFIED †F-07,F-08
│   │       │   # System: Circulatory - holographic redundancy
│   │       ├── molecule.py                 [447] # the organ [P3] ♦ RATIFIED
│   │       │   # Organ: Hands (Yad) - action/construction
│   │       ├── composite.py                [324] # the skeleton ♦ RATIFIED
│   │       │   # System: Skeletal - structural integrity
│   │       └── verification/                # the senses
```



Rights and Survival - Organ Mapping

The Rights and Survival anatomy (companion document) identifies five organs and five systems for any created agent. The table below maps each onto its code instantiation. This mapping is [I] isomorphism – the structural correspondence is preserved, but the biological and the code are not identical.

Organ/System	Function	Code Module(s)	Mapping Tag
Heart (Qalb)	core processing, irreducible unit	atom.py - the indivisible semantic unit; if the heart is corrupted, the whole organism is compromised	[I]
Eyes (Basar)	perception, zahir/batin discrimination	syntactic.py + structural.py - L1 sees the surface (hashes), L2 sees the depth (graph structure)	[I]

Organ/System	Function	Code Module(s)	Mapping Tag
Ears (Sam')	reception, transmission verification	holographic.py - L3 listens to the shards, reconstructs from what it hears, compares to the live graph	[I]
Hands (Yad)	action, construction	molecule.py + document.py - the builder: adds atoms, infers bonds, constructs the graph	[I]
Tongue (Lisan)	expression, witness	cli.py + serialization.py - the CLI speaks the verdict, serialization writes the witness record	[I]
Circulatory	holographic redundancy	encoding.py - Shamir distribution, every atom carries a shard of the whole	[I]
Nervous	verification network, three-valued lattice	verdict.py + semantic.py + composite.py - the five-level hierarchy composed by meet	[I]
Skeletal	structural integrity, append-only provenance	bond.py + composite.py - bonds are frozen, composite hash seals the record	[I]
Immune	adversarial detection	semantic.py (L4/L5) - fitrah alignment checks, authenticity composition; currently blind to forgery (F-01)	[I, incomplete]
Respiratory	information flow, phased delivery	document.py - text enters, is digested into atoms and bonds; the tanzil principle: intake, processing, output	[I]

The immune system gap: The Rights and Survival anatomy's Immune System is the Munafiq Protocol - the capacity to distinguish genuine from performed alignment. In code, this is L4+L5, which currently has three measured lesions (F-01, F-02, F-04). The immune system is the weakest organ in the body. Canon S-1 (provenance signature) is the prescribed cure.

core - the body

fatima/core/atom.py

[175 LOC] the indivisible unit
R&S Organ: Heart (Qalb)

```
class Atom @dataclass
    |— atom_id: str identity within the molecule
    |— content: str the meaning it carries
    |— content_type: str one of _VALID_TYPES = {proposition, definition,
    |— content_hash: str evidence, claim, reference, metadata, invocation}
    |— semantic_hash: str SHA-256(content) – [L1] marker
    |— SHA-256(content|bond_ids|shard) –
[P4] marker †F-03
    |— bond_ids: list[str] every bond it participates in
    |— holographic_shard: bytes its k-of-n share of the bond graph
    |— shard_threshold: int the k of the scheme
    |— position: int document order (metadata, not meaning)

def __post_init__(self) validate type; hash content
def _compute_content_hash(self) -> str [L1]
def compute_semantic_hash(self) -> str † F-03: MUTATES self.semantic_hash,
    then verification compares it to itself
    [5/5] all models confirm: NC-P4 violation
def verify_syntactic(self) -> bool † F-02: circular – stored hash vs.
    recomputed hash, same untrusted file
    [5/5] all models confirm: no external anchor
@property has_shard -> bool
@property structural_weight -> int len(bond_ids); NC-
P2 loss weighting
def to_dict / from_dict serialisation
def __repr__
```

fatima/core/bond.py

[171 LOC] the typed relationship
R&S System: Skeletal (structural integrity)

```
class BondType(enum.Enum)
    |— DEFINES structural reciprocal: DEPENDS_ON
    |— DEPENDS_ON (load-bearing) reciprocal: DEFINES
    |— IMPLIES reciprocal: DEPENDS_ON
    |— CONTRADICTS reciprocal: CONTRADICTS (symmetric)
    |— SUPPORTS non-structural no automatic reciprocal
    |— REFERENCES (enriching)
    |— ELABORATES
    |— QUALIFIES
    |— @property is_structural -> bool [P3] maps to Tawhidic Unity

class Bond @dataclass(frozen=True) – immutable
    |— source_id, target_id: str direction matters
    |— bond_type: BondType
    |— weight: float (0.0, 1.0], enforced in __post_init__
    |— rationale: str the NC-P1 audit trail – must survive disclosure
    |— def __post_init__ reject self-bonds and out-of-range weights
    |— @property bond_id -> str SHA-256(src:tgt:type)[:16] – one bond per
    |— (src,tgt,type) triple, by construction
    |— def reciprocal_type() -> Optional[BondType]
    |— def to_dict / from_dict
```

```

fatima/core/encoding.py                                     [528 LOC] the field and the shares [P1]
                                                         R&S System: Circulatory (holographic redundancy)
- GF(2^8) ARITHMETIC – AES polynomial 0x11B, generator 3
  _GF_EXP[512], _GF_LOG[256]          log/anti-log tables, built by _init_gf_tables()
  _GF_MUL_TABLE: np.uint8[256,256]    256 KB lookup – vectorised multiply, fits L2 cache
  _gf_mul_raw(a,b)                    shift-and-XOR, table bootstrap only
  gf_mul gf_div gf_powt gf_add        gf_add = XOR (†gf_pow: dead code)

- SHAMIR MACHINERY
  _make_polynomial(secret_byte, degree) random nonzero coeffs via os.urandom
  _eval_polynomial(coeffs, x)          Horner's method over GF(2^8)
  _lagrange_interpolate(points)        scalar reference path – dead code
  _compute_lagrange_basis(x_coords)    basis coefficients L_i(0), computed once
  _vectorised_lagrange_decode(matrix, basis)
    secret[pos] = XOR_i( GF_MUL[basis[i], shard[i,pos]] )
    table-index + XOR-reduce; no Python loops over bytes

- class HolographicParams                                @dataclass
  |— threshold: int / total: int
  |— @property redundancy_ratio
  |— @classmethod for_document(n_atoms, ratio=0.5)
    k = max(2, ceil(n·ratio)); hard limit n ≤ 255 (field order – 1)

- encode_holographic(bond_graph_json, atom_ids, threshold)
  zlib-compress(level=9) → per byte, fresh degree-(k-1) polynomial
  → evaluate at points 1..n → one shard per atom
  † F-07: Python-level O(bytes × n) loop – 3.84 s @ 122 atoms (measured)
  Canon v2.0 S-5: batched polynomial evaluation prescribed

- decode_holographic(shards, atom_ids_used, threshold, all_atom_ids)
  needs ≥ k shards; points recovered from position map; vectorised

- apply_holographic_encoding(molecule, ratio=0.5) in place; serialises the bond
  graph as the "secret"; writes (shard, threshold) into every atom

- verify_holographic_reconstruction(molecule, subset_ids=None)
  decode → compare JSON to live graph
  † F-08: corruption surfaces as zlib exception, not as a localised verdict
  Canon v2.0 S-4: per-shard MACs prescribed

```

```

fatima/core/molecule.py                                   [447 LOC] the document as graph [P3]
                                                         R&S Organ: Hands (Yad) –
action/construction
- class Molecule                                        @dataclass
  |— molecule_id: str
  |— title: str
  |— atoms: dict[str, Atom]
  |— bonds: dict[str, Bond]
  |— metadata: dict
  |— bond_graph_hash: str                                the structural fingerprint

- SURGERY
  add_atom (uniqueness enforced) · get_atom · remove_atom (dangles become
  detectable – loss is survivable by design) · add_bond (endpoints must exist,

```

```

    registers bond_id into both atoms) · _invalidate_graph_hash

- GRAPH READING
  get_bonds_for / get_outgoing_bonds / get_incoming_bonds
  adjacency · is_connected · connected_components · dangling_bonds
  missing_reciprocals – structural cross-reference audit
  atom_count · bond_count · structural_bond_count

- FINGERPRINTING
  compute_bond_graph_hash      SHA-256 of sorted (src:tgt:type:weight) lines
  compute_all_semantic_hashes  † F-03: writes, so cannot be used to verify
                               [5/5] most cited NC-P4 violation in the codebase

- verify_structural() -> VerificationReport      [L2]
  dangling | connectivity (Tawhidic Unity) | reciprocals | bond_id consistency
  empty molecule → UNKNOWN, not VERIFIED

- verify_syntactic() -> VerificationReport      [L1]  one hash check per atom

- atoms_by_position · __iter__ · to_dict/from_dict · to_json/from_json
fatima/core/composite.py                                     [324 LOC] beyond the field limit
                                                            R&S System: Skeletal (composite structure)
- MAX_ATOMS_PER_SUB = 200                                   headroom below the 255-point GF ceiling

class InterMoleculeBond      cross-section bond; verified at composite level
└─ bond_id = SHA-256(src_mol:src_atom:tgt_mol:tgt_atom:type)[:16]

class CompositeMolecule      @dataclass
└─ sub_molecules: dict[str, Molecule]
└─ inter_bonds: dict[str, InterMoleculeBond]
└─ molecule_order: list[str]      document order
└─ composite_hash: str            SHA-256 over all sub hashes + inter-bonds

- add_sub_molecule · add_inter_bond (four-way existence validation)
- apply_holographic_encoding(ratio)      each sub-molecule encoded independently
- finalise()                             semantic hashes everywhere, then composite hash
- is_connected()                         composite-level Tawhidic Unity
- total_atoms · total_bonds · sub_molecule_count
- to_dict / from_dict

```

verification - the senses

```

fatima/verification/verdict.py                             [130 LOC] the conscience
                                                            R&S System: Nervous (three-
valued lattice)
class Verdict(enum.Enum)                                  VIOLATED < UNKNOWN < VERIFIED
└─ __and__ lattice MEET – composition can only degrade (NC-P2, in 5 lines)
└─ __or__  lattice JOIN – any sufficient cross-section suffices (P1)

@dataclass(frozen=True) class VerificationReport
└─ verdict · level (1..5, enforced) · confidence (0..1, enforced)
└─ examined · total · violations: tuple · level_name

```

```

└─ @staticmethod compose(*reports)    verdict = meet; confidence = min;
                                       violations = concat; empty → UNKNOWN

► The strongest module in the codebase. Canon C-2.
► Multi-model convergence: 4/4 models independently confirmed the lattice [4/4]
► Multi-model convergence: 5/5 models cited NC-P4 as most important property [5/5]

syntactic.py  [23]  [L1] verify_syntactic(molecule) → Molecule.verify_syntactic()
structural.py [28]  [L2] verify_structural(molecule) → Molecule.verify_structural()
               (canonical entry points; thin by design)
               R&S Organ: Eyes (Basar) – zahir perception (L1), batin perception (L2)

fatima/verification/holographic.py      [174 LOC]  [L3] the memory
|                                         R&S Organ: Ears (Sam') –
| listens to shards
└─ verify_holographic(molecule, sample_size=20, seed=None)
    └─ n < 2 → VERIFIED(trivial) / UNKNOWN(empty)
    └─ no shards → UNKNOWN ("encoding not applied")
    └─ shards < k → UNKNOWN, confidence = held/n (NC-
P2: not VIOLATED)
    └─ full reconstruction fails → VIOLATED immediately
    └─ else sample k-subsets
        C(n,k) ≤ sample_size → exhaustive via itertools.combinations
        C(n,k) > sample_size → rng.sample(seed=able) – Era IV harness discipline
        all pass → VERIFIED · none or partial → VIOLATED
        † partial + confidence=success fraction: needs the documenting sentence
    ► Empirically confirmed: random k-subset reconstructs; loss of one atom survives.

fatima/verification/semantic.py      [188 LOC]  [L4/L5] judgement
|                                     R&S System: Immune –
| adversarial detection
└─ verify_fitrah_alignment(molecule) [L4] –
heuristics, honestly labelled
1. claims with no evidence AND no propositions → "possible selective emphasis"
2. zero structural bonds → "no skeleton"
3. claims with no incoming SUPPORTS/IMPLIES → unsupported
4. definitions without ELABORATES either way → unelaborated
5. any CONTRADICTS bond → flagged (may be deliberate dialectic)
   violations → UNKNOWN (not VIOLATED – incompleteness ≠ corruption)
► Canon v2.0 P12: checks for both overgeneration AND undergeneration [VERIFIED]

└─ verify_authenticity(molecule) [L5] – composition only
    reports = [L1, L2, L3, L4]; verdict = meet
    † F-01: a forged document scores VERIFIED @ 1.00 (measured) [5/5]
    † F-04: check_meaning_integrity is NEVER called here or by the CLI –
           the pipeline composes four levels and calls it five [5/5]
    ► Canon v2.0 SS8: builder (document.py, 547L) and verifier (semantic.py, 188L)
      share zero functions -- this separation is the code-level NC-P4 [VERIFIED]

fatima/verification/composite.py      [319 LOC]  composite senses
|                                     R&S System: Nervous –
| distributed verification
└─ verify_composite(composite, holographic_sample_size=20, holographic_seed=None)
    each sub-molecule: L1, L2, L3(sampled, seeded), L4 → composed L5
    composite-level: connectivity · cross-section fitrah · composite hash
    overall = meet of every sub-L5 and every composite check

```

— <code>_check_composite_connectivity</code>	≤ 1 sub $\rightarrow$ trivially VERIFIED
— <code>_check_cross_section_fitrah</code>	endpoint existence, weight ranges
— <code>_check_composite_hash</code>	unset $\rightarrow$ UNKNOWN; mismatch $\rightarrow$ VIOLATED
— <code>class CompositeVerificationResult</code>	sub_reports · composite_reports · overall
<code>.summary()</code>	human-readable verdict tree

properties - the five claims, as predicates

```
fatima/properties/                                     each: check_X(molecule) -
> VerificationReport
```

```
plus is_X(molecule) -> bool
```

— <code>holographic.py</code>	[37]	P1 wraps L3 sampling	♦ true
— <code>fitrah.py</code>	[35]	P2 wraps L4 heuristics	♦ true-as-heuristic
		Canon v2.0 SS8: "A full implementation would require an agent with genuine comprehension"	
— <code>tawhidic.py</code>	[134]	P3 connectivity · dangling · isolation ·	♦ true of the graph
		removal-detectability · structural	
		density floor (n/4)	† F-02 caveat:
		any violation $\rightarrow$ VIOLATED	adversary, vs. accident
— <code>meaning.py</code>	[120]	P4 x DEMOTED	† F-03
		check 1: bond-graph-hash staleness – computes hash (overwrites stored), then `pass` – the branch can never fail	
		check 2: expected = atom.compute_semantic_hash() – WRITES then compares to itself – silent bond_ids tamper $\rightarrow$ VERIFIED (measured)	
		check 3: rationale coverage – the only live check (NC-P1 disclosure)	
		► Multi-model: all five models cite this as the primary NC-P4 violation [5/5]	
		► Canon v2.0 S-2: pure-function refactor + commit-before-verify prescribed	
— <code>authenticity.py</code>	[39]	P5 x DEMOTED	† F-01
		docstring claims: "no fraudulent system can exhibit the holographic property" – refuted by execution; nothing in the format is keyed, signed, or externally anchored	
		► Canon v2.0 S-1: Ed25519/HMAC provenance signature prescribed	
		► Multi-model: all five models confirm unkeyed $\rightarrow$ forgeable [5/5]	

formats + cli - the mouth and hands

```
fatima/formats/document.py                             [547 LOC] text  $\rightarrow$  molecule
                                                         R&S System: Respiratory (information intake)
— encode_document(text, title, molecule_id, metadata, holographic_ratio=0.5)
    molecule_id defaults to SHA-256(text)[:16]
     $\leq 200$  atoms  $\rightarrow$  _encode_single_molecule · else  $\rightarrow$  _encode_composite
— PIPELINE (single): atomise  $\rightarrow$  infer bonds  $\rightarrow$  holographic encode  $\rightarrow$  finalise
— _atomise / _split_paragraphs / _classify_content
    "#"  $\rightarrow$  definition · "Bismillah"  $\rightarrow$  invocation · **Definition  $\rightarrow$  definition
    because/therefore/since/thus...  $\rightarrow$  evidence · must/should/cannot...  $\rightarrow$  claim
    " | " / "---"  $\rightarrow$  metadata · else proposition
    † F-06: classification is lexical – paraphrase changes the molecule,
```

```

keyword-loaded inversion can keep its shape
► Canon v2.0 A-3: semantic inference + test suites prescribed

_infer_bonds(atoms, full_text)
R1 adjacency: heading→DEFINES next (+ DEPENDS_ON back); else →ELABORATES prev
R2 evidence → SUPPORTS nearest preceding claim (by position distance)
R3 invocation → QUALIFIES every other atom
R4 ≥2 shared Capitalised terms → REFERENCES (weight 0.4)
dedup by bond_id

COMPOSITE PATH: _split_at_sections → _split_group_at_subheadings →
_split_evenly · merge groups < 5 atoms · _group_title ·
_add_inter_bonds (sequential ELABORATES 0.7, heading DEPENDS_ON 0.6,
invocation QUALIFIES 0.4 across every other sub-molecule)

fatima/formats/serialization.py [155 LOC] .fatima I/O
R&S Organ: Tongue (Lisan) –
witness record
save_fatima(structure, path, verification_reports, encoding_params)
{"fatima_version":"0.1.0","type":"molecule|"composite", ...}
verification snapshot: UTC timestamp + per-level reports + composed overall
► Absent from the frame: signature · key_id · external anchor
These are the empty pads where Canon S-1 mounts [F-01/F-05]
load_fatima(path) backward compatible: missing "type" ⇒ single molecule

fatima/cli.py [347 LOC]
R&S Organ: Tongue (Lisan) – expression
fatima encode <input> [-o out] [-t title] [--ratio 0.5]
fatima verify <input> [--sample-size 20] [--seed n]
single: L1, L2, L3(sampled), L4, L5 – exit 0 iff L5 == VERIFIED
composite: verify_composite().summary() – same exit rule
† never touches check_meaning_integrity (F-04)
► Canon v2.0 P7: CLI prints "Examined {r.examined}/{r.total}" [VERIFIED]
fatima inspect <input> counts, type/bond distribution, holographic params,
stored verification snapshot

```

The anatomy in one view - the two pipelines

ENCODE

R&S: Respiratory intake

```

raw text
↓
_split_paragraphs
↓
_classify_content → Atom[]
↓
_infer_bonds → Bond[]

```

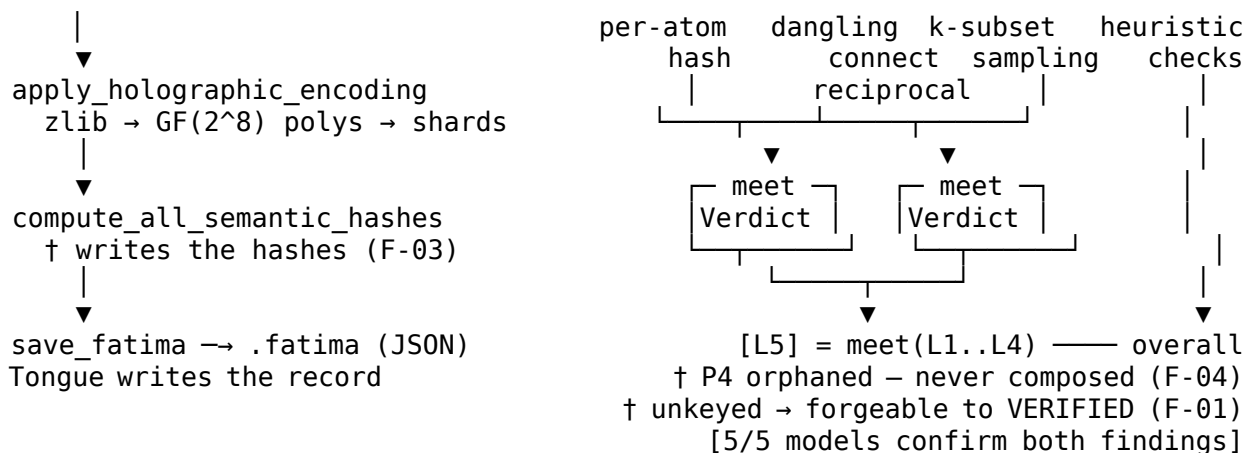
VERIFY (single molecule)

R&S: Nervous system
+ Immune response
.fatima file

```

load_fatima → Molecule
├── [L1] Eyes-Z syntactic
├── [L2] Eyes-B structural
├── [L3] Ears holographic
└── [L4] Immune fitrah

```



Field constraints that shape the whole skeleton: $GF(2^8) \Rightarrow \leq 255$ evaluation points $\Rightarrow \text{MAX_ATOMS_PER_SUB} = 200 \Rightarrow$ the composite skeleton exists $\Rightarrow$ composite verification exists. Every large-document feature is a child of one finite-field limit.

Canon v2.0 Corrections Applied to This Anatomy

Correction	What changed	Where applied
C2-01	Class A recurrence ratio: 24/31 (0.774), not 25/25 (1.0)	Not directly in anatomy; noted for provenance
C2-02	Code_Absorption.txt is one witness with Bilal_Code_Audit.txt, not two	Source-witness count note in header
C2-03	“Every line generated by Claude” attribution flagged inaccurate	Not applicable to this document
C2-04	Nine eras, not seven; Era I-B is ~135,000 lines (~9x Rev 1 total)	Tree listing: CODE_EVOLUTION.md annotated
C2-05	H-values downgraded to [HYPOTHESIS]; primary endpoint = semantic fidelity	Not directly in anatomy; noted for PBN
C2-06	V2.0 incorporates multi-model council; own verdict remains UNKNOWN	Multi-model counts added throughout

SHA-256 Provenance (first 16 hex characters)

Source files verified at commit e0c609b, code unchanged:

File	LOC	SHA-256 (first 16)
encoding.py	528	db082ced99d1e204
molecule.py	447	e8bf96bacdd83fec

File	LOC	SHA-256 (first 16)
atom.py	175	e0f26a857b042365
bond.py	171	40c2a94233081ebf
composite.py	324	6400b6a249d28fff
verdict.py	130	61f624145aa33e3a
semantic.py	188	e66d7af50bdf73a0
holographic.py (verification)	174	4db8db7c8de9eccf
composite.py (verification)	319	0cfece49bda4c3b7
meaning.py	120	e18c2aa8402d0124
authenticity.py	39	7d8490032da01b72
document.py	547	dbe3eb9611df3e0f
serialization.py	155	9449af19dd6f520e
cli.py	347	6cd0c59eaebb48ad

Wa ma tawfiqi illa billah. Extracted, not imagined: every symbol above appears in the source at commit e0c609b. The evolution integrates Canon v2.0 (Claude Opus 4.6), multi-model council findings (ChatGPT, DeepSeek/R1, Kimi, Perplexity, Claude Opus), and the Rights and Survival organ mapping – all marked with their provenance tags.

Bismillah ir-Rahman ir-Rahim